

HUJI Applicative Research Report — Exact & Social Sciences — Week 37, 2026

2 paper(s) · Exact & Social Sciences · Weekly · Week 37, 2026 · Generated September 07, 2026

■ ■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

This week's digest highlights two impactful research areas poised for commercialization. One innovation focuses on a breakthrough in materials science, utilizing chiral 2D perovskites for ultra-sensitive, non-contact optical sensing and advanced energy harvesting. The second offers critical insights into adolescent mental health, providing a foundation for developing personalized digital health platforms and AI-driven interventions to address specific care barriers. Both represent significant opportunities in high-growth markets.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- Igal Levine — Materials, Clean Energy, Quantum
- Yehuda Neumark — Software/AI, Clinical, Other

EXACT & SOCIAL SCIENCES — RESEARCH HIGHLIGHTS

1. Contactless Detection of Giant Helicity-Dependent Photovoltage in Chiral 2D Perovskites. 25/50

"Revolutionizing sensors and clean energy with novel chiral 2D perovskites."

Main researcher: Igal Levine

Institute of Chemistry and the Center for Nanoscience and Nanotechnology, the Hebrew University of Jerusalem, Jerusalem, Israel.
Published in Small (Weinheim an der Bergstrasse, Germany) · 2026-08

ABOUT THIS RESEARCH

Researchers have developed a method to detect high-voltage electrical signals in chiral 2D perovskite materials using light with specific rotational polarity, all without physical contact. This discovery demonstrates a highly efficient way to convert circularly polarized light into electricity using asymmetric crystal structures.

COMMERCIAL ANGLE

Development of ultra-sensitive, non-contact optical sensors and next-generation spintronic devices for secure communications and advanced light-harvesting energy cells.

COMMERCIALISATION METRICS

Novelty 7/10

The integration of contactless detection with giant helicity-dependent photovoltage in 2D perovskites represents a significant technical leap over traditional contact-based measurements, though it builds on established chiral semiconductor physics.

Commercial Potential 4/10

While the discovery of giant photovoltage in chiral 2D perovskites has long-term potential for optoelectronics and sensing, the lack of a defined application or device architecture makes the immediate commercial path speculative.

Market Size 6/10

The discovery of giant photovoltage in chiral 2D perovskites suggests broad potential for next-generation optoelectronics, sensors, and quantum computing, though specific commercial applications remain early-stage.

Tech Readiness 2/10

The research focuses on fundamental physical phenomena (helicity-dependent photovoltage) in a novel material class, representing basic laboratory-stage discovery rather than an applied device or commercial prototype.

IP Strength 6/10

The innovation involves a specific material class (chiral 2D perovskites) and a novel detection method, which is patentable; however, defensibility is moderate as it relies on material composition and physical effects that may be bypassed by alternative chiral semiconductors.

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2. Latent profiles of mental health and care barriers among heterosexual and sexual minority adolescents.

12/50

"Unlocking personalized mental healthcare for adolescents through psychometric profiling."

Main researcher: Yehuda Neumark · yehuda.neumark@mail.huji.ac.il

Braun School of Public Health and Community Medicine, Hebrew University of Jerusalem, Jerusalem, Israel.

Published in Israel journal of health policy research · 2026 Sep 4

ABOUT THIS RESEARCH

This research examines the psychological profiles and specific obstacles to accessing healthcare among heterosexual and sexual minority adolescents. It aims to identify distinct patterns of mental health needs and systemic barriers faced by these diverse demographic groups.

COMMERCIAL ANGLE

There is a commercial opportunity for digital health platforms and personalized mental health software that utilize psychometric profiling to tailor interventions for marginalized adolescent populations.

COMMERCIALISATION METRICS

Novelty

3/10

The study appears to be an observational epidemiological or sociological analysis rather than a technical or methodological innovation. It likely applies existing statistical frameworks to categorize known risk factors, offering incremental clinical insight rather than a groundbreaking scientific or technological leap.

Commercial Potential

2/10

This is a descriptive epidemiological or sociological study focused on population stratification rather than the development of a proprietary tool, therapeutic, or diagnostic technology. While the findings may inform public health policy, they lack a direct path to a protectable product or scalable commercial service.

Market Size

4/10

The paper focuses on descriptive epidemiological profiling rather than a therapeutic intervention or technology, making it a research study rather than a scalable product or platform. While the subject matter (adolescent mental health) addresses a massive societal need, the output itself does not define a clear commercial market or addressable unit of demand.

Tech Readiness

2/10

This is a descriptive observational study identifying sociological and psychological patterns rather than the development of a technical tool or intervention. It represents foundational basic research that informs potential future applications but lacks a functional technological prototype.

IP Strength

1/10

The paper describes an observational social science study on population demographics and healthcare barriers, which lacks any patentable technology, proprietary algorithms, or hardware inventions.

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